

# Data Card: Agricultural Extension RAG Retrieval Benchmark

## Dataset Summary

|                                    |                                                         |
|------------------------------------|---------------------------------------------------------|
| <b>Name</b>                        | Agricultural Extension RAG: Smart Retrieval for Farmers |
| <b>Task type</b>                   | Document retrieval / ranking (TREC / BEIR-style)        |
| <b>Domain</b>                      | Agriculture — Sub-Saharan Africa                        |
| <b>Language</b>                    | English                                                 |
| <b>Corpus size</b>                 | 695 documents                                           |
| <b>Train queries</b>               | 308 (labeled)                                           |
| <b>Test queries</b>                | 200 (labels hidden)                                     |
| <b>Relevance judgments (train)</b> | 4,194 graded qrels                                      |
| <b>Metric</b>                      | nDCG@5                                                  |
| <b>Baseline</b>                    | TF-IDF, nDCG@5 = 0.551                                  |
| <b>Context</b>                     | Built for the TRI AI Saturdays cohort competition       |

Given a smallholder farmer’s natural-language question (e.g. “*Why are my maize leaves turning yellow?*”), the task is to retrieve and rank the five most relevant documents from a fixed agricultural-extension knowledge base. It represents the retrieval half of a RAG pipeline — the component that decides which passages an LLM should read before generating an answer.

## Motivation

- **Purpose:** Provide a realistic, intent-sensitive retrieval benchmark for agricultural-extension RAG systems serving smallholder farmers, rather than a generic open-domain QA/retrieval set.
- **Gap addressed:** Farmer questions frequently share surface vocabulary (crop names, symptom words) across genuinely different problems and intents. A benchmark built purely on lexical overlap would reward keyword matching rather than the disambiguation an extension chatbot actually needs. The graded relevance scheme and explicit hard negatives are designed to test that.
- **Creators/funders:** Built internally for the TRI AI Saturdays cohort; not attributed to a named research group in the provided materials.

## Composition

### Corpus (documents.csv) — 695 documents

| Field       | Type | Description                                  |
|-------------|------|----------------------------------------------|
| document_id | str  | Unique ID, used as DocumentId in submissions |

|            |     |                                                                              |
|------------|-----|------------------------------------------------------------------------------|
| title      | str | Document title                                                               |
| text       | str | Body — a short extension factsheet                                           |
| source     | str | Attributed publisher style (FAO, CGIAR, Plantwise, IITA, ICRISAT, AGRA, ...) |
| crop       | str | Primary crop, or (general) for soil/climate topics                           |
| country    | str | Example country context                                                      |
| origin     | str | synthetic or llm_grounded                                                    |
| source_url | str | Source link (grounded docs only)                                             |
| license    | str | License of underlying source (CC-BY for grounded docs)                       |

Topic coverage: crop diseases, pests, nutrient deficiencies, soil, climate, fertiliser.

## Queries and labels

| File                    | Rows  | Description                                                                       |
|-------------------------|-------|-----------------------------------------------------------------------------------|
| train_queries.csv       | 308   | query_id, query, positive_docs (space-separated doc IDs with relevance $\geq 1$ ) |
| test_queries.csv        | 200   | query_id (starts at 1001), query — no answers provided                            |
| qrels_train.csv         | 4,194 | query_id, document_id, relevance $\in \{0, 1, 2, 3\}$                             |
| sample_submission.csv   | 1,000 | Exact submission format (200 queries $\times$ 5 rows)                             |
| baseline_submission.csv | 1,000 | Working TF-IDF baseline output                                                    |

## Relevance scale (graded, not binary)

| Score | Meaning                                                                                                                                                                |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3     | Perfect — directly answers the question (e.g. the identification facet of nitrogen deficiency in maize, incl. agro-zone variants)                                      |
| 2     | Relevant — a genuinely complementary facet of the same topic (e.g. symptoms supporting identification). Right crop + problem but wrong intent does <i>not</i> qualify. |
| 1     | Marginal — same issue, different crop                                                                                                                                  |

|   |                                                                                                                                                                       |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0 | Not relevant — includes hard negatives that are lexically similar but answer the wrong intent (e.g. prevention vs. causes), plus documents about a confusable problem |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|

The 0-labeled hard negatives are a deliberate design choice: they exist specifically to separate intent-aware retrieval from keyword matching.

## Collection / Construction Process

- **Provenance split:** Documents are either
  - `synthetic` — original content authored for this competition, or
  - `llm_grounded` — derived from CC-BY/CC0 open-access materials (chiefly CGIAR/CGSpace) via LLM rewriting.
- **Generation process:** Produced by an internal pipeline that caches all source fetches and LLM rewrites, making dataset generation deterministic/reproducible.
- **No stated human-annotation process** for the qrels beyond the graded scheme above (the card does not describe annotator identity, count, or inter-annotator agreement — treat this as an open question if annotation quality needs to be audited).

## Licensing & Attribution

| Origin       | License                               | Attribution requirement                                                                                                                                                     |
|--------------|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| synthetic    | CC0 (public domain)                   | None. The source field is a <i>stylistic</i> attribution to the type of organization that publishes such guidance — it is <b>not</b> a claim the text was copied from them. |
| llm_grounded | Inherits source license (CC-BY / CC0) | Must retain attribution via <code>source_url</code> and <code>license</code> columns. NoDerivatives (ND) and NonCommercial (NC) sources were excluded during filtering.     |

When reusing individual documents, check the `license` and `source_url` columns per-row rather than assuming a blanket license for the corpus.

## Intended Uses

- Education and benchmarking of retrieval / RAG systems.
- Built specifically for the TRI AI Saturdays cohort competition.

## Out-of-Scope / Prohibited Uses

- **Not agronomic advice.** Synthetic and LLM-rewritten text may contain simplifications or factual errors and must not be used to

inform real farming decisions.

- Not representative of regions or farming systems outside Sub-Saharan Africa — geographic generalization claims should not be made from this corpus.

## Known Limitations

- **Synthetic/LLM-rewritten content risk:** A portion of the corpus is LLM-generated or LLM-rewritten, which can introduce subtle inaccuracies not present in the original grounded sources.
- **Annotation transparency:** The qrels' annotation methodology (who labeled, how consistency was checked) is not documented in the source materials.
- **Domain narrowness:** 695 documents is a small, closed corpus — good for controlled benchmarking, but not representative of the long tail of real-world extension knowledge.
- **English-only:** No multilingual or low-resource-language variants are included, despite the target user base (smallholder farmers) often being more comfortable in local languages.

## Evaluation Protocol

- **Metric:** nDCG@5, applied identically to both Public and Private leaderboards; the two leaderboards differ only in which held-out queries they score against.
- **Why nDCG@5:** graded (non-binary) relevance and multiple relevant docs per query mean the metric needs to reward both *finding* relevant documents and *ranking* the most relevant ones first — nDCG@5 captures both.
- **Submission format:** long-format CSV, exactly 5 rows per test QueryId, row order = predicted rank (1st row = rank 1). 1,000 rows total for 200 test queries. No score column.

```
QueryId,DocumentId
1001,42
1001,87
1001,19
1001,55
1001,12
```

- **Baseline to beat:** TF-IDF + bigrams  
(TfidfVectorizer(stop\_words="english", ngram\_range=(1,2), min\_df=2)) with cosine similarity scores **nDCG@5 = 0.551** — clearly above random but leaves headroom for semantic/hybrid retrievers to close the gap on the intent-sensitive hard negatives.

## Maintenance

- No stated update cadence, versioning scheme, or maintainer contact in the source materials — treat the dataset as a static, fixed release for this competition unless told otherwise.